, Pune

MIT-ADT UNIVERSITY
PUNE, INDIA
A Leap Towards World Class Education

Name: Ipshita Mahadwaue
Subject: Prog. language Class: SOC-8 Batch: A
Roll No.: 18
Name of the Experiment: Assignment-No. 1 Exp. No.:
Performed on:
Submitted on:

Marks	Teacher's Signature with date
A 10	8/8

Assignment Number ~ 1.

Ques.1 List the key feature and advantages of Python. Also mention any two Python IDEs or editor used for program development.

Ans: Interpreted language - Program are executed line by line, which makes debugging easier.
High level language - No need to manage memory explicity; Python handles it automatically
Open source - Free to use and distribute
Extensive library support - Comes with a large standard library for tasks like file handling, web development.

Advantages.

1. Fast development - Less code is required compared to other language like C or Java.
2. Easy debugging and Maintenance - Simple syntax reduces and makes program easier to maintain

Incomplete for Theory ☐ Diagrams ☐ Observation Table ☐
Calculations ☐ Graphs ☐ Results ☐ Conclusion ☐
Understanding Through Q/A ☐ Late Submission ☐ Neatness ☐

Teacher's Signature

5. Versatile - Used in web development, machine learning, AI, data science, automation, and more.

Any two python IDES

- Pycharm
- Jypur notebook

2. Explain the history and evolution of python. Why is python popular for biginners and industry.

History & evolution

- Python was created by Guido Van Rossum in 1989.
- Released in 1991
- Named after Monty Python
- Designed for simplicity and readability.
- Python 3x is the latest and widely used version.

Why python is popular.

- Easy syntax for biginners.
- used in AI, ML, web development, Data science.
- Huge community support.
- used by companies like google, Netfix, instagram.

faster development with fewer lines of code

*. Python Program to accept 2 numbers and perform arithmatic operation ?

```
a = float (input ("Enter first number :"))
b = float (input ("Enter second number :"))

print (" Addition ", a+b )
print (" Substraction =", a-b)
print ( " Multiplication ," a*b)
print (" Division:" , a/b )
```

Output

```
Enter first number : 10
Enter second number : 5
Addition : 15
Substraction : 5
Multiplication = 50
division : 2.0
```

4. Analyze the following python code and identify the data type of each variable. Also explain the output.

```
a = 10
b = 5.5
c = 3+4j
d = True
print (a+b, type (c), d )
```

Data types.

- a → int
- b → float
- c → complex
- d → bool

Output explanation

- a+b - 15.5
- type (c) → (class 'complex'>
- d → true.

5. Evaluate the use of relational, logical and bitwise operators in python. Justify with suitable example where each type of operator is most appropriate.

Relational operator: used for comparison, used in conditional and decision making.

a = 10
b = 5
print (a>b)

- logical operators
used to combine conditions
used in authentication and validation

x = True
y = false
print (x and y)

- Bitwise operators
operate on binary numbers
used in low level prog. and networking

a = 5
b = 3
print (a & b)

6. Design and implement a menu-driven python program that:
- accepts user input
- Performs arithmetic, relational, logical and assignment operators.
- Display result clearly.

The program should handle int, float and boolean data types.

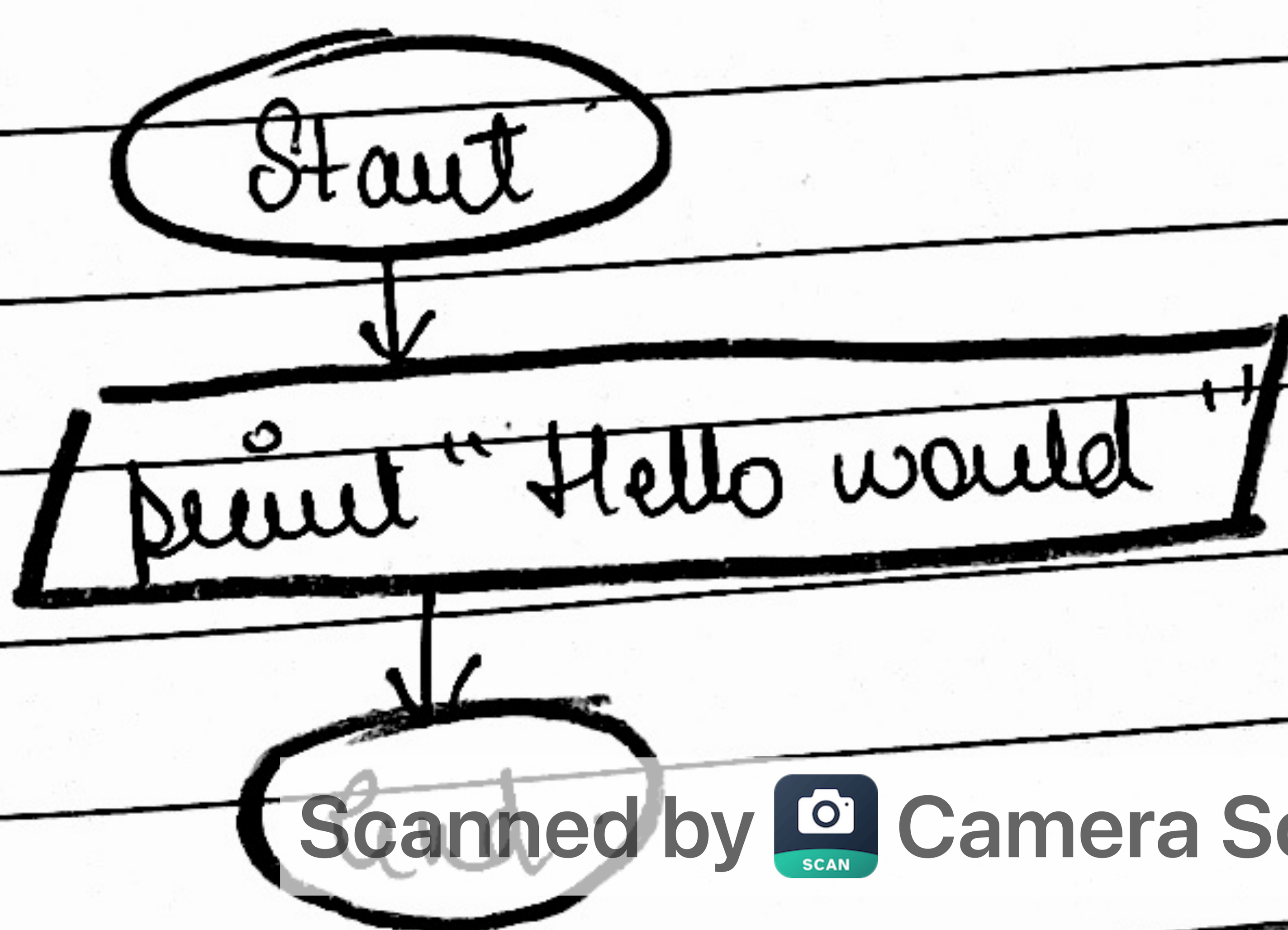
```
a = int (input (" Enter first number : "))
b = int (input (" Enter second number : "))

print ("1. Arithematic")
print ("2. Relational")
print ("3. logical")
print (" 4. Assignment ")

ch = int( input (" Enter choice : "))

if  ch = = 1 :
    print ( " Add = ", a+b)
elif
    ch  =  2
    print (" a>b = ", a > b)
el if
    ch  = 4 :
    a + = b
    print ("a = ", a)
else
    print (" invalid choice ")
```

7. Write a program in python to display "Hello world"



Code

```
print ("Hello world")
```

Output

Hello world

Algorithm

1. Start
2. Print Hello World.
3. Stop.

8. Write a program to perform average of 5 numbers using input () function and type conversion concept.

```
a = float (input ("Enter number 1: "))
b = float (input (" Enter number 2: "))
c = float (input (" Enter number 3: "))
d = float (input (" Enter number 4 : "))
e = float( input (" Enter number 5 :"))

Avg = (a + b + c + d + e) | 5
print (" average : ", avg)
```

Output.

Enter number 1 = 10

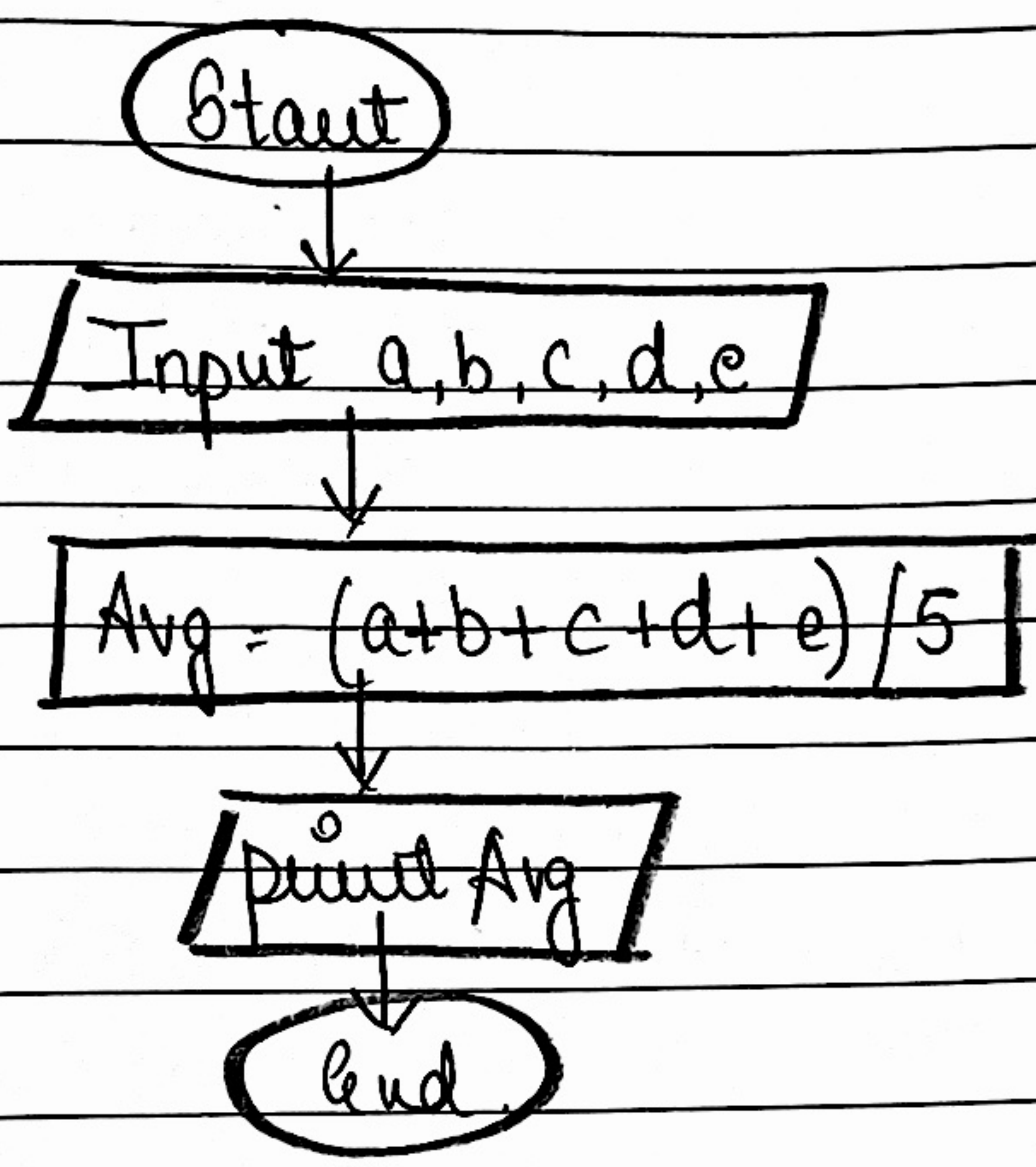
Enter number 2 = 20

Enter number 3 = 30

Enter number 4 : 40
Enter number 5 : 50
Average = 30.0

Algorithm

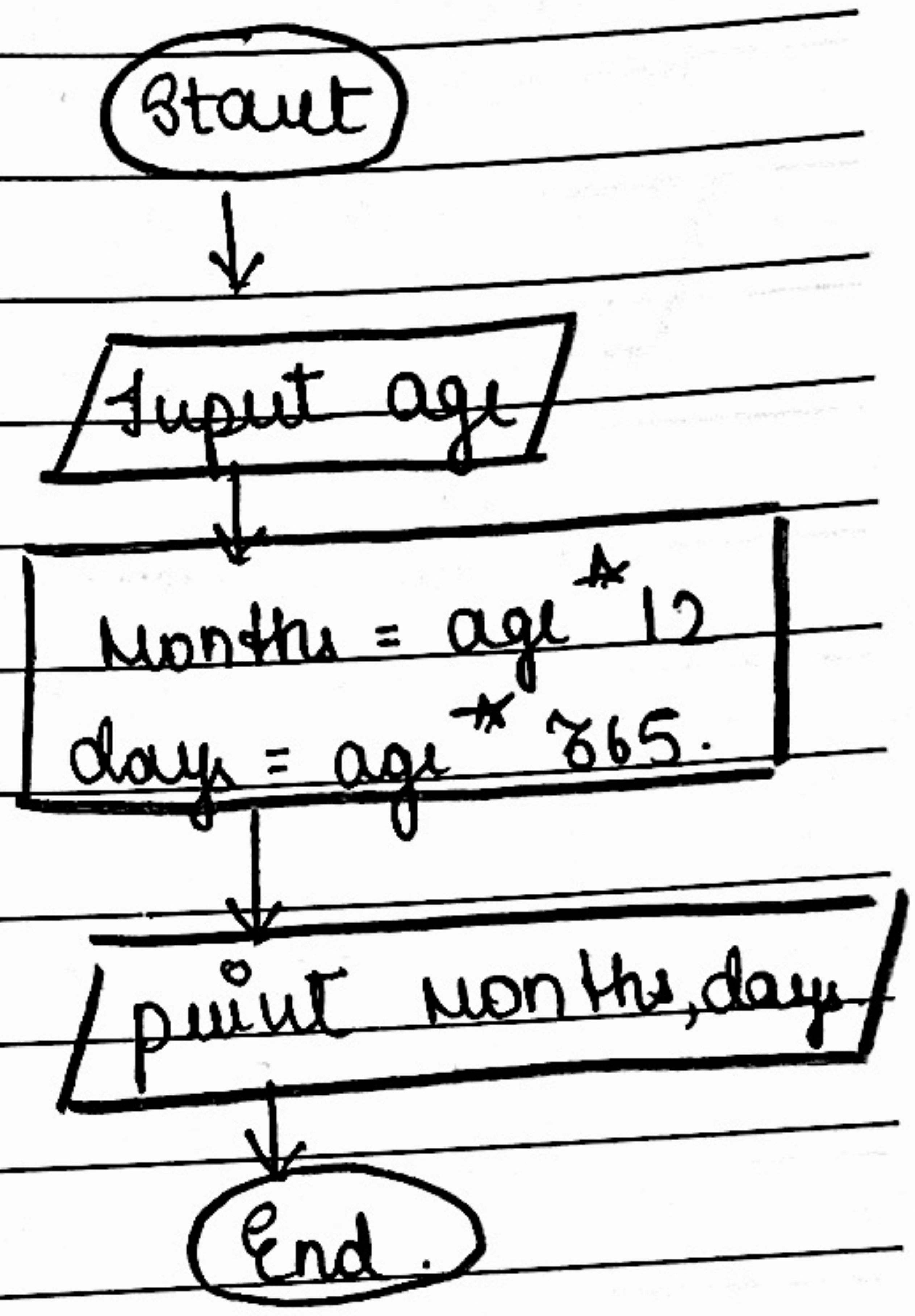
1. Start
2. Accept 5 numbers from user.
3. Calculate average.
4. Display average
5. Stop



Que.9. Write a program that promts the user to enter their age and prints out their age in years, months and days.

```
age = int (input ("Enter age in years : "))
months = age * 12
days = age * 365.
print (" Age in months : ", months)
print (" Age in days : ", days)
```

Output



Enter Age in years : 18
Age in Months : 216
Age in days : 6 570

Algorithm

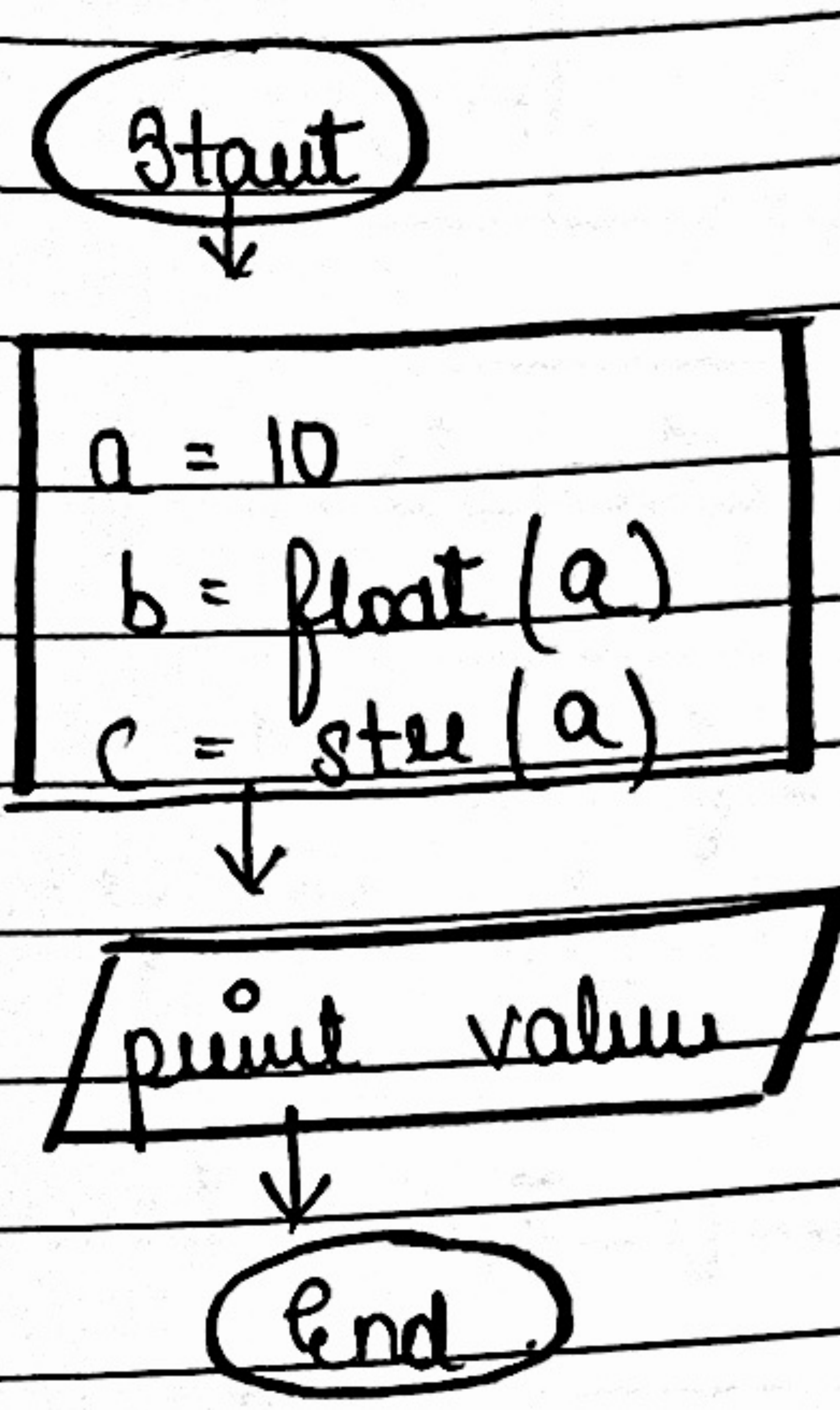
1. Start
2. Input age in years.
3. calculate Months = age x 12
4. Calculate days = age x 365
5. Display month and days
6. Stop

Jun. 10

Write a program to demonstrate the concept of data types & conversion

```
a = 10
b = float(a)
c = str(a)

print (a, type(a))
print (b, type(b))
print (c, type(c))
```



Output

10 <class 'int'>
10.0 <class 'float'>
10 <class 'str'>

Algorithm

1. Start
2. Assign integer value.
3. Convert integer to float and string
4. Display values and their data types.
5. Stop.